

S V V Sai Sri Vallabh Pataneni

Research Assistant/Associate (PhD): Fuel Cell Systems, PEM & SOFC

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SUMMARY

A highly motivated Automotive Engineering Master's student with a deep passion for hydrogen mobility. Possesses a comprehensive understanding of **fuel cell system control**, gained by architecting and building the complete **cathode-side control architecture**, including a **Model Predictive Control (MPC)** approach, for the ~150-200 kW EnginePod aircraft powertrain at DLR. Combines this system-level control expertise with hands-on leadership of **Germany's only student-built hydrogen fuel cell vehicle** at Ecogenium e.V. and industrial FCEV operating-strategy experience at Robert Bosch GmbH, backed by a peer-reviewed manuscript in review at SAE. Eager to bring this fuel cell systems expertise, from control architecture to test-bench validation, to the **Research Assistant/Associate PhD position** at RWTH's Chair of Thermodynamics of Mobile Energy Conversion Systems (TME).

EXPERIENCE

Werkstudent, Fuel Cell Systems (Aircraft), DLR Institute for Small Aircraft Technologies (INK) 01/2026 – Present

Würselen (Aachen), Germany

- **Cathode-Side Control Architecture:** Architected and built the complete cathode-side control architecture for the ~150-200 kW EnginePod fuel cell system powering DLR's hydrogen-electric 9-seater commuter aircraft, developing a 0D lumped-parameter plant model in MATLAB/Simulink and layering feed-forward, PI, decoupler and Model Predictive Control (MPC) strategies on top, now calibrated against real system measurement data.
- **Higher-Fidelity Modeling:** Building a complementary 1D lumped-parameter plant model of the cathode system in GT Suite to validate and extend the MATLAB/Simulink control architecture.
- **Balance-of-Plant Sizing & Selection:** Built Python- and Excel-based sizing tools to select balance-of-plant components, deriving design specifications for the cathode, thermal-management and sensory systems and driving their procurement.
- **System Rig Design:** Designed the complete fuel cell system rig in CATIA V5 for packaging, piping, testing and component orientation, optimized for easy transferability and integration.

Operating Strategy / System Simulation Intern, Robert Bosch GmbH 09/2025 – 12/2025

Schwieberdingen, Germany

- **Operating-Strategy Modelling:** Developed a MATLAB/Simulink model for an existing power-split operating strategy concept and integrated it into the (FC)EVsim PHEV module, validating its fidelity by correlating simulation outputs against the original developer's reference Excel and measurement data.
- **Degradation-Safeguarding Strategy:** Engineered a new, elaborate operating strategy in a MATLAB script for Strong Fuel Cell Hybrid EVs, focused on safeguarding fuel cell and HV-battery degradation by managing dynamic loads, thermal stress, low-voltage operation, SOC swing and life cycles, parameterized by MAPs for different battery capacities and drive cycles.
- **Multi-Objective Optimization:** Conducted multi-variable parametric optimization to generate the required MAPs, specifically targeting charge sustenance and minimal fuel consumption.
- **Analysis & Publication:** Developed post-processing plots to analyze efficiency and degradation KPIs, benchmarked the new strategy's performance against Dynamic Programming (DP) and ECMS, and co-authored a peer-reviewed manuscript on the strategy, now in review at SAE.

STUDENT CLUB

Ecogenium e.V, Mechanical / Suspension Team Lead & Pit Crew 06/2024 – Present

Aachen, Germany

- **Fuel Cell System Testing & Troubleshooting:** Leading this season's fuel cell system testing, integration and fault diagnosis on Germany's only student-built hydrogen fuel cell vehicle, on its newly built water-cooled stack, running structured test campaigns across the electrical, thermal and fluidic subsystems.
- **Energy Management & Cooling Strategy:** Designed and deployed the energy-management and cooling-system control strategy for the new water-cooled fuel cell system in MATLAB/Simulink.
- **Fuel Cell Test Bench Development:** Engineered, fabricated, and electronically integrated a standalone fuel cell test bench for the team's prior air-cooled stack, executing operational testing (purging, humidity cycling, short-circuiting) and performance data evaluation, and mapped the stack's optimal thermal operating window.
- **Strategy Implementation:** Architected and deployed advanced driving and fuel cell temperature-control strategies using Simulink; leveraged real-time competition data to boost fuel efficiency by 80%, a critical factor in securing a runner-up finish at the Shell Eco-Marathon.

EDUCATION

M.Sc. Automotive Engineering, RWTH Aachen University

09/2023 – present

Aachen, Germany · GPA 1.7 (1.2 in core automotive courses) · Master's thesis planned at DLR on fuel-cell system control · Relevant modules: Alternative & Mobile Propulsion Systems, Simulation Sciences, Mechatronics, Thermodynamics, FEM

B.Tech Mechanical Engineering, JNTU Hyderabad

06/2019 – 04/2023

Hyderabad, India · CGPA 9.05 (GPA 1.3, German scale)

SKILLS

Fuel Cell Systems & Controls

PEM fuel cell system architecture

0D/1D lumped-parameter plant modeling

feed-forward/PI/decoupler control design

Model Predictive Control (MPC)

back-pressure-valve (BPV) protection control

operating-strategy optimization

balance-of-plant sizing & selection

CAN communication

Modeling & Simulation

MATLAB/Simulink

GT Suite

Dymola (Modelica)

Genetic Algorithms

Gradient Descent

DP/ECMS benchmarking

Test & Measurement

fuel cell test-bench build & commissioning

single-cell & stack testing

polarisation curves

purging/humidity cycling

measurement-data evaluation & troubleshooting

Programming & Data

Python (NumPy, Pandas)

C/C++

Power BI

Design & CFD

CATIA V5

Siemens NX

ANSYS (Fluent, Structural)

LANGUAGES

English (Fluent, C1) · German (Intermediate, B1) · Telugu / Hindi (Native)

RECOGNITIONS AND PUBLICATIONS

Paper Publication, IJSED Research

"A Comprehensive Analysis of Two Innovative Aircraft Design Configurations"

Paper Publication, Robert Bosch GmbH (in review, SAE)

Co-author, rule-based FCEV operating-strategy control approach